



BAULKHAM HILLS HIGH SCHOOL

JULY 2014
YEAR 11 TASK 2

Mathematics

General Instructions

- Working time – 60 minutes, plus 5 minutes reading time
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 6 to 14
- Marks may be deducted for careless or badly arranged work

Total marks – 47

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Page 2

Questions 1-5 (5 marks)

- Attempt Question 1-5

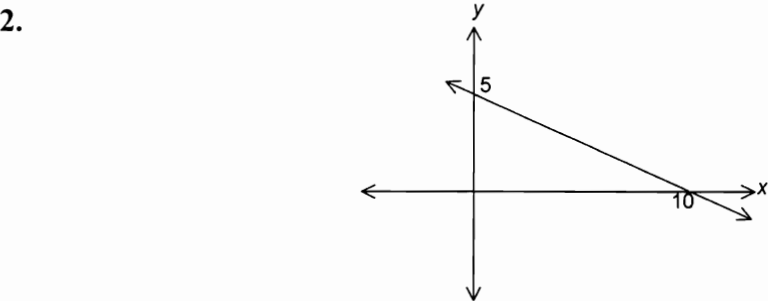
Section II – Pages 3-5 (42 marks)

- Attempt questions 6 to 14

Section I - 5 marks
Attempt questions 1-5

Use the multiple choice answer sheet (in the booklet) for question 1-5

1. The midpoint of the line joining $(0,-5)$ and $(d,0)$ is:
(A) $\left(\frac{d-5}{2}, 0\right)$ (B) $\left(0, \frac{5-d}{2}\right)$ (C) $\left(\frac{d}{2}, \frac{-5}{2}\right)$ (D) $\left(\frac{d}{2}, \frac{5}{2}\right)$



- The angle of inclination with the positive x-axis of the line shown above is approximately:
- (A) 27° (B) 63° (C) 117° (D) 153°
3. ABCD is a parallelogram with diagonals AC and BD.
Consider the following statements:

- I. If these diagonals are perpendicular, then ABCD is a rhombus.
II. If these diagonals are equal, then ABCD is a square.

- Which of these statements are correct?
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
4. The centre of a circle is $(-3,4)$ and the circle passes through $(1,2)$. The length of the diameter is:
(A) $2\sqrt{2}$ (B) $2\sqrt{5}$ (C) $4\sqrt{2}$ (D) $4\sqrt{5}$

5. Evaluate $\lim_{x \rightarrow \infty} \frac{3x^3 - 4x + 5}{x^2 - 3x^3}$
(A) 1 (B) -1 (C) 3 (D) -3

Section II – Extended Response

Attempt questions 6-14.

Answer each question in the booklet provided. Clearly indicate question number.

All necessary working should be shown in every question.

6.

Find the equation of the line which is perpendicular to $y = -2x + 3$ and passes through $(3,-4)$. Answer in general form.

3
7.

Evaluate, clearly showing working to justify your answer:
$$\lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$$

2
8.

Differentiate the following

a) $y = 3x^4 + 4x - 5$

2

b) $y = \frac{x^4 - 10x^5}{5x^2}$

2

c) $f(x) = (x^2 + 1) \cdot \sqrt{x - 1}$

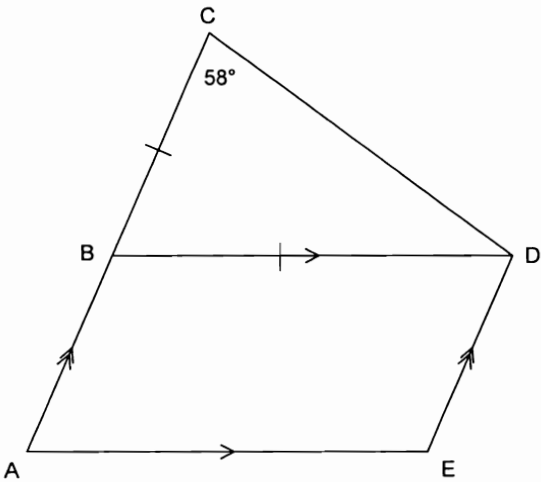
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d) $y = \frac{x+1}{4x-1}$

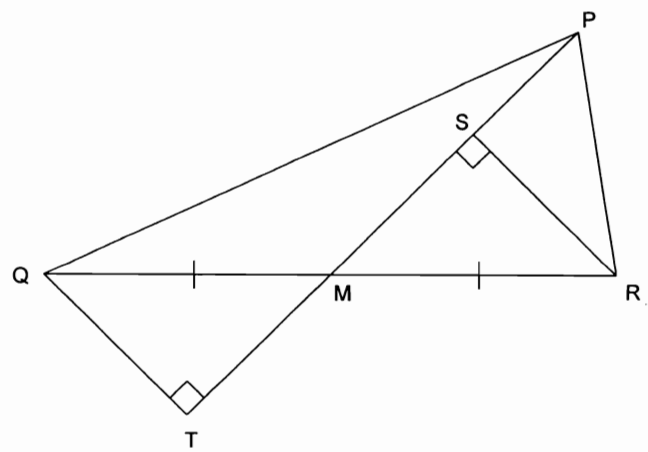
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9.

In the diagram shown, $AB \parallel ED$, $BD \parallel AE$, $BC = BD$ and the points A, B and C are collinear. If $\angle BCD = 58^\circ$, find the size of $\angle DEA$, giving all reasons

3

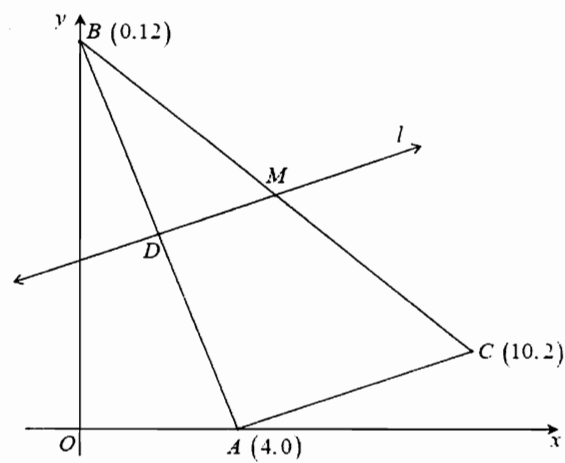


10. QT and RS are perpendicular to PT. M is the midpoint of QR and TS=15cm.



- a) Prove that $\triangle QMT$ is congruent to $\triangle RMS$. 3
- b) Find the length of MT with reasons. 1
- c) What type of quadrilateral is QSRT? Justify your answer. 1

11.



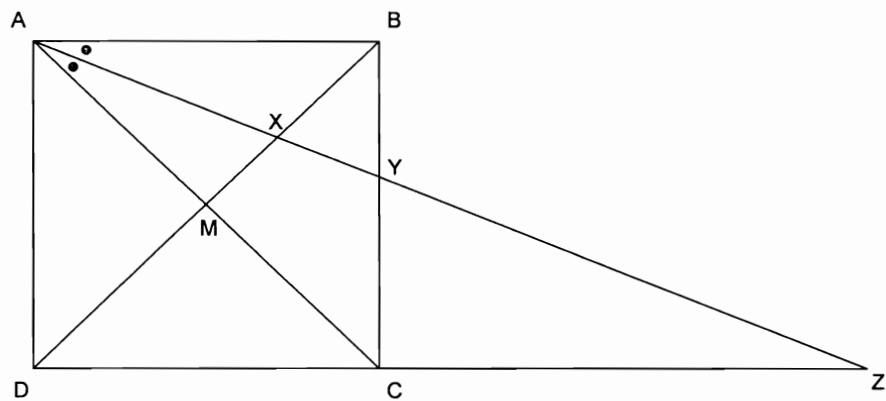
In the above diagram, A, B and C are the points (4,0), (0,12) and (10,2) respectively

- (i) Find the gradient of AC. 1
- (ii) Find the coordinates of D, the midpoint of AB. 1
- (iii) The line l is parallel to AC and passes through D. Show that the equation of l is $x - 3y + 16 = 0$ 1
- (iv) The line l meets BC at M. Explain why M must be the midpoint of BC. 1
- (v) Find the perpendicular distance from C to line l . 2
- (vi) Find the area of $\triangle DMC$. 2

12. Find the derivative of $f(x) = 4x - x^2$ by first principles. 3

13. Find the equation of the tangent to the curve $y = (x^2 + 1)^3$ at the point on the curve where $x = 1$. 3

14. 3



ABCD is a square whose diagonals intersect at M. The bisector of $\angle BAC$ cuts BM at X and BC at Y. DC and AY are produced to meet at Z.

- a) Prove $\triangle AMX$ is similar to $\triangle ZCY$. 3
- b) Find the ratio of CY:MX, giving reasons. 4

$$1. \left(\frac{0+d}{2}, \frac{-5+0}{2} \right) \\ = \left(\frac{d}{2}, -\frac{5}{2} \right) \quad \textcircled{C}$$

$$2. m = -\frac{1}{2} = \tan \theta \\ \theta \div 153^\circ \quad \textcircled{D}$$

$$3. \text{I true, II false.} \quad \textcircled{A}$$

$$4. r = \sqrt{(1+3)^2 + (2-4)^2} \\ = \sqrt{20} \\ = 2\sqrt{5} \therefore d = 4\sqrt{5} \quad \textcircled{D}$$

$$5. \lim_{x \rightarrow \infty} \frac{3 - \frac{4}{x^2} + \frac{5}{x^3}}{\frac{1}{x} - 3} \\ = \frac{3-0+0}{0-3} = -1 \quad \textcircled{B}$$

$$6. y = -2x+3 \text{ has } m = -2 \\ \text{Perp. } m = -\frac{1}{-2} = \frac{1}{2} \\ y+4 = \frac{1}{2}(x-3) \\ 2y+8 = x-3 \\ x-2y-11 = 0$$

$$7. \lim_{x \rightarrow 3} \frac{(x-3)(x^2+3x+9)}{(x-3)(x+3)} \\ = \frac{9+9+9}{3+3} \\ = \frac{27}{6} \text{ or } 4.5$$

$$8. a) y = 3x^4 + 4x - 5 \\ y' = 12x^3 + 4$$

$$b) y = \frac{x^4 - 10x^5}{5x^2} = \frac{1}{5}(x^2 - 10x^3)$$

$$y' = \frac{1}{5}(2x - 30x^2) \leftarrow \\ \text{or equivalent.}$$

$$c) f(x) = (x^2+1) \cdot \sqrt{x-1} \\ u = x^2+1 \quad u' = 2x \\ v = \sqrt{x-1} \quad v' = \frac{1}{2}(x-1)^{-1/2} \cdot 1 \\ = \frac{1}{2\sqrt{x-1}}$$

$$f'(x) = uv' + vu' \\ = (x^2+1) \cdot \frac{1}{2\sqrt{x-1}} + \sqrt{x-1} \cdot 2x$$

$$d) y = \frac{(x^2+1)(x-1)^{1/2} + (x+1)^{1/2} \cdot 2}{4x-1}$$

$$u = x+1 \quad u' = 1 \\ v = 4x-1 \quad v' = 4$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{(4x-1) \cdot 1 - (x+1) \cdot 4}{(4x-1)^2}$$

$$= \frac{4x-1-4x-4}{(4x-1)^2}$$

$$= \frac{-5}{(4x-1)^2}$$

$$9. \angle BDC = 58^\circ \quad (\text{Equal } \angle \text{s opposite equal sides of } \triangle) \\ \angle ABD = 2 \times 58^\circ \quad (\text{Exterior } \angle \text{ of } \triangle BCD \\ = 116^\circ \quad (\text{Sum of interior opposite } \angle \text{s}))$$

$$ABCD \text{ is a parallelogram (Opposite sides parallel)} \\ \therefore \angle DEA = 116^\circ \quad (\text{opposite } \angle \text{s of parallelogram equal})$$

$$10. a) \text{ In } \triangle QMT, \triangle RMS:$$

$$\angle QTM = \angle SRM \quad (\text{both } 90^\circ \text{ or given})$$

$$\angle QMT = \angle SMR \quad (\text{vertically opposite } \angle \text{s})$$

$$QM = MR \quad (\text{given})$$

$$\therefore \triangle QMT \cong \triangle RMS \quad (\text{AAS})$$

$$b) MT = \frac{1}{2} \times 15 = 7.5 \text{ cm}$$

$$(M \text{ is the midpoint of } TS)$$

$$c) QRST \text{ is a parallelogram (diagonals bisect each other).}$$

$$11. i) m_{AC} = \frac{2-0}{10-4} = \frac{2}{6} = \frac{1}{3}$$

$$ii) D = \left(\frac{0+4}{2}, \frac{12+0}{2} \right) = (2, 6)$$

$$iii) \text{ Through } (2, 6) \text{ with } m = \frac{1}{3}:$$

$$y - 6 = \frac{1}{3}(x - 2)$$

$$3y - 18 = x - 2$$

$$x - 3y + 16 = 0$$

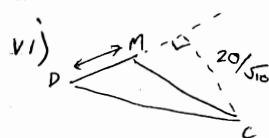
iv) Intercepts on parallel lines are proportional
OR Parallel lines cut all transversals in the same ratio.
Line ~~joining~~ ^{from} midpoint of one side of a $\triangle$,
parallel to another side, meets the third side
at its midpoint

v) $C(10, 2)$ Line $l: x - 3y + 16 = 0$

$$d = \left| \frac{(10) - 3(2) + 16}{\sqrt{1^2 + (-3)^2}} \right| \quad \leftarrow 1$$

$$= \left| \frac{20}{\sqrt{10}} \right|$$

$$= \frac{20}{\sqrt{10}}$$



$P(2, 6)$ $M = (5, 7)$... midpt of BC

$$DM = \sqrt{(5-2)^2 + (7-6)^2}$$

$$= \sqrt{9 + 1}$$

$$= \sqrt{10}$$

$$\therefore \text{Area} = \frac{1}{2}bh = \frac{1}{2} \cdot \sqrt{10} \cdot \frac{20}{\sqrt{10}}$$

$$= 10 \text{ units}^2 \quad \leftarrow 1$$

12. $y = f(x) = 4x - x^2$

$$f(x+h) = 4(x+h) - (x+h)^2$$

$$= 4x + 4h - (x^2 + 2xh + h^2)$$

$$= 4x + 4h - x^2 - 2xh - h^2 \quad \leftarrow 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4x + 4h - x^2 - 2xh - h^2 - 4x + x^2}{h} \quad \leftarrow 1$$

$$= \lim_{h \rightarrow 0} \frac{4h - 2xh - h^2}{h}$$

$$= \lim_{h \rightarrow 0} 4 - 2x - h$$

$$= \underline{4 - 2x}$$

* Structure +
notation important

13. $y = (x^2 + 1)^3$

$$y' = 3(x^2 + 1)^2 \cdot 2x$$

$$= 6x(x^2 + 1)^2$$

$$= 6(1+1)^2 \text{ at } x=1$$

$$= 24$$

Pt of contact $(1, 8)$ $(1^2 + 1)^3$

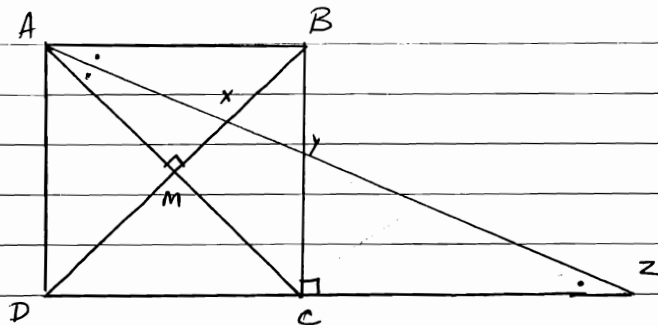
$$y - 8 = 24(x - 1)$$

$$\underline{y = 24x - 16}$$

14.

14.

13.



a) $\angle AMX = 90^\circ$ (diagonals of square perpendicular)

$\angle BCD = 90^\circ$ (Ls of square are right Ls)

$\therefore \angle ZCY = 90^\circ$ ($BC \perp DC$)

$\therefore \angle AMX = \angle ZCY$ (both 90°)

$\angle MAX = \angle BAX$ (AX bisects $\angle BAC$)

$AB \parallel DC$ (opposite sides of square parallel)

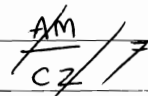
$\angle BAX = \angle CZY$ (alternate Ls, $AB \parallel DC$)

$\therefore \angle MAX = \angle CZY$ (both = $\angle BAX$)

$\therefore \triangle AMX \parallel \triangle ZCY$ (matching Ls equal)

b) $\triangle CZA$ is isosceles (2 equal Ls from (a))

$\therefore CA = CZ$ (Equal sides opposite equal Ls of $\triangle CZA$)



$\frac{CY}{MX} = \frac{CZ}{AM}$ (matching sides of similar $\triangle$ s proportional)

$= \frac{CA}{AM}$ ($CA = CZ$ from above)

$= \frac{2 \cdot AM}{AM}$ (diagonals of square bisect each other)

(*) logical reasoning leading to

(*)